# Supplementary Material for "Mitigating State-Action Distribution Shifts in Offline Reinforcement Learning via Implicit State-State Planning"

## Appendix A. Proof for Theorem 1

*Proof*: Consider two distinct Q-functions, $\hat{Q}(s,a)$ and $\hat{Q}'(s,a)$. According to the definition of $\hat{\mathcal{B}}^{\pi_g}$, its infinite norm operation satisfies

$$
\begin{aligned}
&\left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q})(s,a) - (\hat{\mathcal{B}}^{\pi_g}\hat{Q}')(s,a)\right\|_\infty \\
=&\max_{s,a}\left|(r+\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}(s',a'_{IS})]) - (r+\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}'(s',a'_{IS})])\right| \\
\leq&\max_{s'}\left|\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}(s',a'_{IS}) - \hat{Q}'(s',a'_{IS})]\right| \\
\leq&\gamma\max_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}\left|\hat{Q}(s',a'_{IS}) - \hat{Q}'(s',a'_{IS})\right| \\
\leq&\gamma\max_{s,a}\left|\hat{Q}(s,a) - \hat{Q}'(s,a)\right| \\
=&\gamma\left\|\hat{Q}(s,a) - \hat{Q}'(s,a)\right\|_\infty.
\end{aligned}
\tag{A.1}
$$

According to Eq. (A.1), $\hat{\mathcal{B}}^{\pi_g}$ is a $\gamma$-contraction mapping, which satisfies $\left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q}) - (\hat{\mathcal{B}}^{\pi_g}\hat{Q}')\right\|_\infty \leq \gamma\left\|\hat{Q} - \hat{Q}'\right\|_\infty$. Setting $\hat{Q}^*$ as the unique fixed point of the Q-function $\hat{Q}$, then repeatedly performing $\hat{\mathcal{B}}^{\pi_g}$ yields

$$
\begin{aligned}
\left\|\hat{Q}_{k+1} - \hat{Q}^*\right\|_\infty &= \left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q}_k) - (\hat{\mathcal{B}}^{\pi_g}\hat{Q}^*)\right\|_\infty \\
&\leq \gamma\left\|\hat{Q}_k - \hat{Q}^*\right\|_\infty \leq \cdots \leq \gamma^{k+1}\left\|\hat{Q}_0 - \hat{Q}^*\right\|_\infty.
\end{aligned}
\tag{A.2}
$$

Eq. (A.2) indicates that for $\gamma < 1$, as the number of iterations $k$ increases, $\hat{Q}_{k+1}$ converges to $\hat{Q}^*$, i.e., $\lim_{k\to\infty}\hat{Q}_{k+1} = \hat{Q}^*$. Thus, Theorem 1 is proofed. ■

## Appendix B. Proof for Theorem 2

*Proof*: Under Assumption 1, solving the deterministic MDP through explicit state-stitching can be regarded as solving a sub-MDP via explicit action-stitching. In this sub-MDP, only the mappings $\mathcal{I}(s, s')$ are included, where $\mathcal{I}(s, s')$ returns a set of feasible actions for each state $s$, this is precisely the mechanism we refer to as implicit state-stitching. Specifically, for any given state $s$, we have

$$\hat{Q}(s, a) \geq \max_{s'} \hat{Q}(s, s') = \max_{s'} \hat{Q}(s, a = \mathcal{I}(s, s')) \quad \text{for least one } a. \tag{B.1}$$

Intuitively, this means that there exists at least one action $a$ for which the $Q$-value in the original MDP is no smaller than the maximal $\hat{Q}$-value obtained by stitching over possible successor states $s'$.

Although the actions generated by $\mathcal{I}(s, s')$ may only constitute a subset of the full action space of the original MDP, we assume that the chosen successor state $s'$ is optimal. Under this assumption, the return associated with any action in the original MDP cannot exceed the return achieved through the stitched action sequence provided by $\mathcal{I}(s, s')$. Formally, for each $s$ we have

$$\hat{Q}(s, a) \leq \max_{s'} \hat{Q}(s, \mathcal{I}(s, s')) \quad \text{for all } a. \tag{B.2}$$

Combining Eqs. (B.1) and (B.2), we conclude that for a deterministic MDP, there exist

$$\max_{a} \hat{Q}(s, a) = \max_{s'} \hat{Q}(s, s') = \max_{s'} \hat{Q}(s, a = \mathcal{I}(s, s')). \tag{B.3}$$

This result demonstrates that, in a deterministic MDP, the maximization over actions $a$ in the original Q-function is equivalent to the maximization over successor states $s'$ in the implicit state-stitching formulation. Hence, Theorem 2 is proven. ■

## Appendix C. Proof for Theorem 3

*Proof*: Recalling Eq. (17) and applying the triangle inequality, we obtain

$$
\begin{aligned}
&\left\|\hat{Q}^{\pi_g}\left(s,\pi_g(s)\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \\
=&\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \\
=&\Big\|(\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right))+(\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)-\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)) \\
&\quad+(\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right))+(\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right))\Big\| \\
\leq&\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\|+\left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)-\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \\
&+\left\|\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\|+\left\|\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\|. \qquad \text{(C.1)}
\end{aligned}
$$

Considering $\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\|$, and given the $K_Q$-Lipschitz as stated in Assumption 2, we can derive

$$
\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\| \leq K_Q\left\|\mathcal{I}(s,\mathcal{X}_g(s))-\pi_\beta(s)\right\|. \qquad \text{(C.2)}
$$

Based on $\pi_\beta(s)=\mathcal{I}(s,s')$ and the $K_I$-Lipschitz as stated in Assumption 2, $\left\|\mathcal{I}(s,\mathcal{X}_g(s))-\pi_\beta(s)\right\| = \left\|\mathcal{I}(s,\mathcal{X}_g(s))-\mathcal{I}(s,s')\right\| \leq K_{\mathcal{I}}\left\|\mathcal{X}_g(s)-s'\right\|$ holds. Recalling the state constraint threshold $\left\|\mathcal{X}_g(s)-s'\right\|\leq\epsilon_g$ from Eq. (12), there exist

$$
\left\|\mathcal{I}(s,\mathcal{X}_g(s))-\pi_\beta(s)\right\| \leq K_{\mathcal{I}}\epsilon_g. \qquad \text{(C.3)}
$$

Combining Eqs. (C.2) and (C.3), we obtain

$$
\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\| \leq K_Q K_{\mathcal{I}}\epsilon_g. \qquad \text{(C.4)}
$$

Next, considering $\left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)-\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\|$, and based on the value gaps under different policies as presented in Lemma 1, it follows that

$$
\left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)-\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \leq \left\|\hat{Q}^{\pi_g}(s,a)-\hat{Q}^{\pi_\beta}(s,a)\right\| \leq R_{\max}\sqrt{2K_{\mathcal{I}}\epsilon_g}/(1-\gamma)^2. \qquad \text{(C.5)}
$$

Due to the estimation deviation between the $\hat{Q}^{\pi_\beta}$ estimated by off-policy expectile re-

gression and the $\bar{Q}^{\pi_\beta}$ estimated by MSE, we define the upper bound of the bias between them as $\delta_\tau$:

$$\left\|\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \leq \left\|\hat{Q}^{\pi_\beta}(s,a)-\bar{Q}^{\pi_\beta}(s,a)\right\| \leq \delta_\tau. \tag{C.6}$$

Furthermore, based on the sampling error result from Lemma 2, we can conclude that

$$\left\|\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \leq \left\|\bar{Q}^{\pi_\beta}(s,a)-Q^{\pi_\beta}(s,a)\right\| \leq \delta_{\mathcal{D}}. \tag{C.7}$$

Finally, substituting Eqs. (C.4)-(C.7) into Eq. (C.1), we can derive

$$\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right)-Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \leq \delta_{\mathcal{D}}+\delta_\tau+K_Q K_{\mathcal{I}}\epsilon_g+R_{\max}\sqrt{2K_{\mathcal{I}}\epsilon_g}/(1-\gamma)^2. \tag{C.8}$$

Thus, Theorem 3 is proofed. ■

## Appendix D. Proof for Theorem 4

*Proof*: In ORL, once the training dataset is fixed, the performance gap $\varepsilon_{\mathcal{D}}$ between the actual global optimal policy $\pi^*$ and the local behavior policy $\pi_\beta$ derived from the finite dataset, becomes fixed:

$$\varepsilon_{\mathcal{D}} = J(\pi^*)-J(\pi_\beta). \tag{D.1}$$

While the performance gap between execution-policy $\pi_e$ and optimal policy $\pi^*$ is $J(\pi^*)-J(\pi_e)$, by applying Eq. (20) to extend $|J(\pi^*)-J(\pi_e)|$ and using the triangle inequality, we obtain

$$\begin{aligned}|J(\pi^*)-J(\pi_e)| &= \left|\left(J(\pi^*)-J(\pi_g)\right)+\left(J(\pi_g)-J(\pi_\beta)\right)+\left(J(\pi_\beta)-J(\pi_e)\right)\right| \\ &\leq \left|J(\pi^*)-J(\pi_g)\right|+\left|J(\pi_g)-J(\pi_\beta)\right|+\left|J(\pi_\beta)-J(\pi_e)\right|. \end{aligned}\tag{D.2}$$

Considering $\left|J(\pi^*) - J(\pi_g)\right|$, based on the definition of $J(\pi)$ in Section 2, Lemma 3, and Eq. (D.1), we can derive

$$\begin{aligned}
\left|J(\pi^*) - J(\pi_g)\right| &= \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi^*}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_g}(s)}\left[r(s)\right]\right| \\
&\leq \frac{1}{1-\gamma}\left|\int_s \left(d^{\pi^*}(s) - d^{\pi_g}(s)\right) r(s)\mathrm{d}s\right| \\
&\leq \frac{R_{\max}}{1-\gamma}\int_s \left|d^{\pi^*}(s) - d^{\pi_g}(s)\right|\mathrm{d}s \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \max_s \left\|\mathcal{X}^*(s) - \mathcal{X}_g(s)\right\| \\
&= \frac{R_{\max}}{1-\gamma} C K_T K_I \max_s \left\|\mathcal{X}^*(s) - s' + s' - \mathcal{X}_g(s)\right\| \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \max_s \left(\left\|\mathcal{X}^*(s) - s'\right\| + \left\|s' - \mathcal{X}_g(s)\right\|\right) \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I (\varepsilon_{\mathcal{D}} + \epsilon_g).
\end{aligned} \tag{D.3}$$

By considering $\left|J(\pi_g) - J(\pi_\beta)\right|$ and applying Lemma 3, we obtain

$$\begin{aligned}
\left|J(\pi_g) - J(\pi_\beta)\right| &= \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_g}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\beta}(s)}\left[r(s)\right]\right| \\
&\leq \frac{1}{1-\gamma}\left|\int_s (d^{\pi_g}(s) - d^{\pi_\beta}(s)) r(s)\mathrm{d}s\right| \\
&\leq \frac{R_{\max}}{1-\gamma}\int_s |d^{\pi_g}(s) - d^{\pi_\beta}(s)|\mathrm{d}s \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \max_s \left\|\mathcal{X}_g(s) - s'\right\| \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \epsilon_g.
\end{aligned} \tag{D.4}$$

Similarly, by considering $\left|J(\pi_\beta) - J(\pi_e)\right|$ and utilizing Lemma 3, we obtain

$$\begin{aligned}
\left|J(\pi_\beta) - J(\pi_e)\right| &= \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\beta}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_e}(s)}\left[r(s)\right]\right| \\
&\leq \frac{1}{1-\gamma}\left|\int_s \left(d^{\pi_\beta}(s) - d^{\pi_e}(s)\right) r(s)\mathrm{d}s\right| \\
&\leq \frac{R_{\max}}{1-\gamma}\int_s \left|d^{\pi_\beta}(s) - d^{\pi_e}(s)\right| \mathrm{d}s \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \max_s \left\|s' - \mathcal{X}_e(s)\right\| \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_I \epsilon_e. \qquad \text{(D.5)}
\end{aligned}$$

Substituting Eqs. (D.3)-(D.5) into Eq. (D.2), the performance lower bound of execution-policy $\pi_e$ is

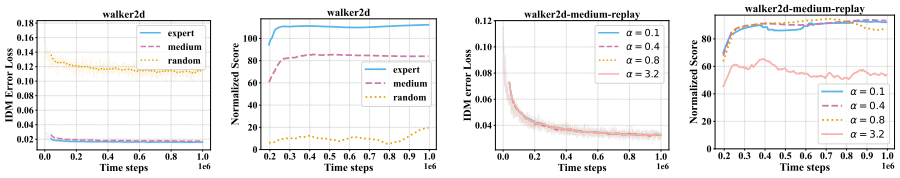
$$\left|J(\pi^*) - J(\pi_e)\right| \leq \frac{R_{\max}}{1-\gamma} C K_T K_I (\varepsilon_{\mathcal{D}} + 2\epsilon_g + \epsilon_e). \qquad \text{(D.6)}$$

Thus, Theorem 4 is proofed.

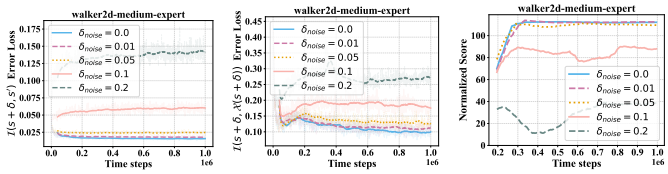
## Appendix E. Reliability Analysis of the Inverse Dynamics Model

Since the IDM transforms STM-generated state transitions into executable actions, its prediction accuracy is critical to the effectiveness of ISSP. Therefore, we further investigate IDM performance under varying dataset qualities, state constraint strengths, and OOD state perturbation intensities. IDM prediction error is quantified by the mean squared error, while the corresponding normalized score is used to evaluate the impact of IDM accuracy on overall policy performance. The results are presented in Fig. E1.

**1) Impact of dataset quality:** Fig. E1(a) demonstrates that higher-quality datasets yield lower IDM prediction errors, with the expert dataset achieving the highest accuracy, followed by the medium and random datasets. Nevertheless, IDM training converges consistently across dataset qualities, indicating that effective inverse dynamics can be learned even from low-quality offline data. The corresponding policy performance exhibits a similar trend, suggesting that accurate inverse dynamics predictions




(a) Varying dataset qualities (b) Varying state constraint strengths




(c) Varying OOD state perturbation intensities

Fig. E1. Impact of dataset quality, state constraint strength, and OOD state perturbation intensity on IDM prediction error and policy performance.

facilitate reliable execution of STM-generated state transitions.

**2) Impact of state constraint strength:** Fig. E1(b) evaluates different values of $\alpha \in \{0.1, 0.4, 0.8, 3.2\}$, where smaller $\alpha$ corresponds to stronger state constraints. The IDM prediction errors remain highly consistent across all $\alpha$ values, indicating that state constraints primarily regulate the transition distribution generated by the STM rather than directly influencing IDM supervised training. In contrast, policy performance varies substantially with $\alpha$, demonstrating that proper state constraints are essential to balance transition generalization and executable action generation.

**3) Impact of OOD state perturbations:** To evaluate IDM robustness beyond the offline state distribution, we perturb the normalized current state as $s_{\text{OOD}} = s + \delta_{\text{noise}}$ with different perturbation intensities $\delta_{\text{noise}} \in \{0, 0.01, 0.05, 0.1, 0.2\}$. Specifically, we consider both $\mathcal{I}(s + \delta_{\text{noise}}, s')$ and $\mathcal{I}(s + \delta_{\text{noise}}, \mathcal{X}(s + \delta_{\text{noise}}))$, where the latter better reflects the complete STM-to-IDM execution pipeline. As shown in Fig. E1(c), IDM prediction error increases gradually with perturbation intensity. Policy performance remains stable under mild perturbations but degrades under larger distribution shifts. These results indicate that the IDM maintains effective state-to-action mapping within

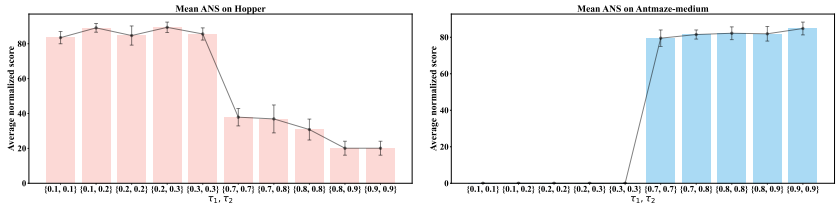
a moderate extrapolation range, whereas substantial OOD shifts impair prediction accuracy and policy performance.

Overall, IDM reliability is jointly determined by dataset quality and the extent of state extrapolation. The state constraint mechanism and state smoothing strategy in Eq. (12) preserve reliable state-to-action correspondence while facilitating moderate generalization beyond the offline dataset.

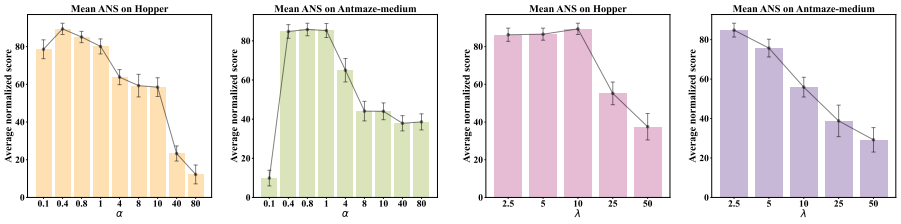
## Appendix F. Hyperparameter Analysis and Selection Insights

To further strengthen the rigor of this study and address **Q6**, we conduct a sensitivity analysis on key hyperparameters: $\{\tau_1, \tau_2\}$, $\alpha$, and $\lambda$. Following standard practice in ORL, these hyperparameters are not tuned separately for each task. Instead, we adopt a category-wise offline tuning protocol, where a single representative environment is used to select hyperparameters that are then fixed for all tasks within the same domain. Specifically, Hopper and AntMaze-medium are chosen as representative environments for dense-reward (MuJoCo/CityLearn) and sparse-reward (AntMaze) domains, respectively. All tuning is conducted strictly offline using the average normalized score on training dataset as the selection criterion. Once chosen, the hyperparameters are applied uniformly across tasks in the corresponding category, as summarized in Fig. F2 and Table G3.

**1) Off-policy Expectile Weight $\{\tau_1, \tau_2\}$:** These parametersjointly control the quantile regression targets of $\hat{Q}$ toward ($\hat{\mathcal{B}}^{\pi_g}\hat{Q}$) during training, thereby strengthening the in-dataset state-action correspondence of STM and amplifying the influence of actions predicted by STM with the aid of IDM on future rewards. Fig. F2(a) shows strong category-level consistency. For the dense-reward Hopper environment, ANS remains stable and high for low expectile weights $\{\tau_1, \tau_2\} < 0.5$. We recommend $\{\tau_1, \tau_2\} \in \{0.2, 0.3\}$, which is applied to all 24 dense-reward tasks. This choice maintains value stability by filtering potential overestimation in high-frequency feedback environments. In contrast, for the sparse-reward AntMaze-medium environment, higher




(a) Off-policy expectile weight $\{\tau_1, \tau_2\}$




(b) Balance coefficient $\alpha$

(c) Mild constraint factor $\lambda$

Fig. F2. Decision-making quality of ISSP under varying off-policy expectile regression weight $\{\tau_1, \tau_2\}$, balance coefficient $\alpha$, and mild constraint factor $\lambda$, on the Hopper and Antmaze-medium environments.

weights $\{\tau_1, \tau_2\} > 0.5$ are necessary to capture rare success signals. Increasing these parameters significantly improves ANS, whereas low values cause failures. We therefore fix $\{\tau_1, \tau_2\} = \{0.9, 0.9\}$ for all 6 sparse-reward tasks. This categorical selection demonstrates that ISSP does not require per-task tuning once the reward density is identified.

**2) Balance Coefficient $\alpha$:** For ISG-PI and MISE-PI, $\alpha$ regulates the trade-off between the mode-seeking objective (maximizing $\hat{Q}$) and the mode-covering property of BCS. Fig. F2(b) shows a consistent trend: ANS initially increases with $\alpha$ but decreases after exceeding a threshold, indicating that a balanced trade-off is essential. We set $\alpha = 0.4$ as a robust default across all domains.

**3) Mild Constraint Factor $\lambda$:** The factor $\lambda$, trained independently within MISE-PI and applied only during inference, regularizes the execution-policy through a mild state mode-covering constraint. Increasing $\lambda$ relaxes this constraint and expands the action exploration space induced by the STM and IDM. Sensitivity analysis in Fig. F2(c) shows that the optimal value is 10 for the Hopper task as well as the dense-

reward MuJoCo/CityLearn domain, and 2.5 for the AntMaze-medium task as well as the sparse-reward AntMaze domain.

## Appendix G. Dataset Description and Hyper-parameter Setting

Table G1. Description and min–max return references for MuJoCo, AntMaze, and CityLearn environments.

| **Control tasks** | **Learning goal** | **Reward** | **Ref. min** | **Ref. max** |
|---|---|---|---|---|
| HalfCheetah | Control a cheetah-like robot to run as quickly forward as possible | Dense | -280.18 | 12135.0 |
| Hopper | Control a monopod-like robot to jump as far forward as possible | Dense | -20.27 | 3234.3 |
| Walker2d | Control a biped-like robot to walk as quickly forward as possible | Dense | 1.63 | 4592.3 |
| Antmaze-umaze | Control an 8-DOF ant-like robot to find the shortest path in a simple-maze layout | Sparse | 0.0 | 1.0 |
| Antmaze-medium | Control an 8-DOF ant-like robot to find the shortest path in a medium-maze layout | Sparse | 0.0 | 1.0 |
| Antmaze-large | Control an 8-DOF ant-like robot to find the shortest path in a difficult-maze layout | Sparse | 0.0 | 1.0 |
| CityLearn | Coordinate domestic hot water and chilled water storage in electricity-consuming buildings to reshape the overall electricity demand profile | Dense | 16280 | 50350 |

Table G2. Dataset types across MuJoCo, AntMaze, and CityLearn environments.

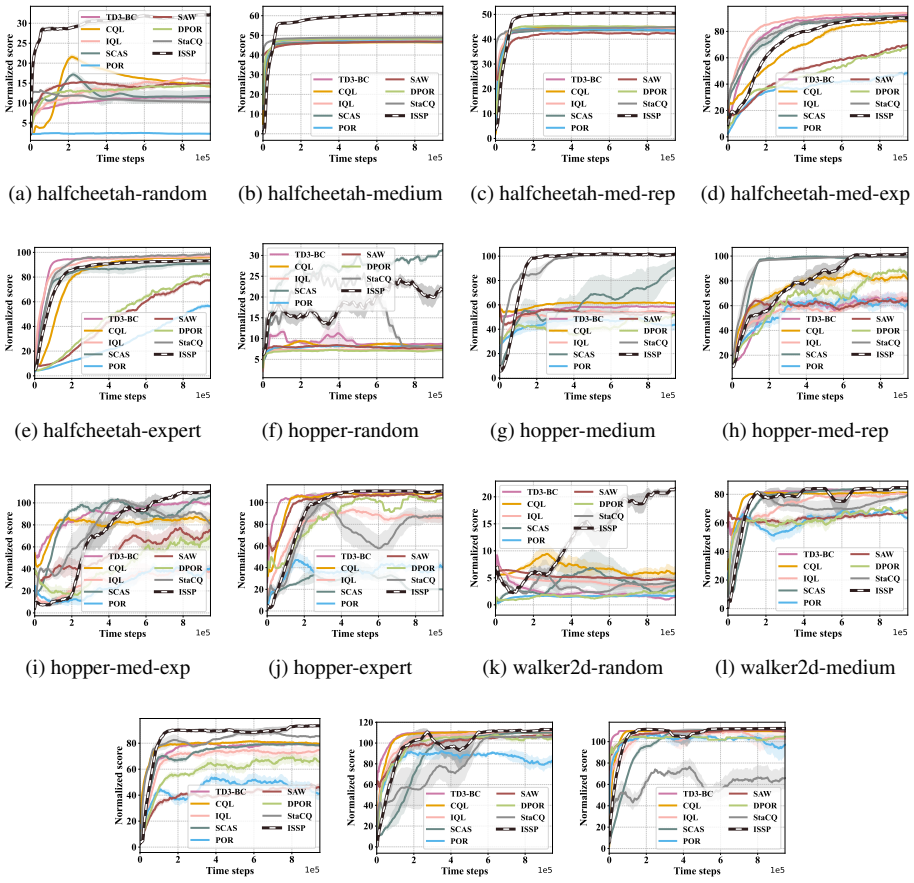
| **Dataset type** | **Description of collected data quality/policy** | **Policy num.** | **Policy mix** | **Experiences** |
|---|---|---|---|---|
| Random | Random-level policy | Single | 100% | $10^6$ |
| Medium | Medium-level policy that achieves one-third of expert policy performance | Single | 100% | $10^6$ |
| Medium-replay | Medium dataset and replay buffer during medium-level agent training | Multiple | 50% : 50% | $\approx 2 \times 10^6$ |
| Medium-expert | Medium-level policy and expert-level policy | Multiple | 50% : 50% | $2 \times 10^6$ |
| Expert | Expert-level agent policy or demonstration | Single | 100% | $10^6$ |
| Default | Goal-reaching policy from a fixed location to a specific goal | Single | 100% | $10^6$ |
| Diverse | Goal-reaching policy from a random location to a random goal | Single | 100% | $10^6$ |
| Play | Starting from hand-picked location set to specific hand-picked goal set | Single | 100% | $10^6$ |
| **Dataset type** | **Description of collected data quality/policy** | **Policy num.** | **Policy mix** | **Trajectories** |
| Low | Collecting policy achieving 25% of expert return, highly suboptimal | Single | 100% | $\{99, 999, 9999\}$ |
| Medium | Collecting policy achieving 50% of expert return, moderately suboptimal | Single | 100% | $\{99, 999, 9999\}$ |
| High | Collecting policy achieving 75% of expert return, near-expert | Single | 100% | $\{99, 999, 9999\}$ |

Table G3. Network structure and hyperparameter setting of ISSP.

| | **Hyper-parameter** | **Value** |
|---|---|---|
| Network | Optimizer | Adam |
| | Mini-batch size | 256 |
| | Activation function | ReLU |
| | Critic/STM/IDM network hidden layer | 4 |
| | Critic/STM/IDM each hidden layer unit | 256 |
| | Critic/STM/IDM network learning rate | $3 \times 10^{-4}$ |
| | All target network update rate $v$ | $5 \times 10^{-3}$ |
| ISSP | Discount factor $\gamma$ | 0.99 |
| | State Exploring noise $\sigma$ | 0.05 |
| | Maximum training time step $T$ | $1 \times 10^{6}$ |
| | STM delay update frequency $d$ | 2 |
| | Off-policy expectile regression weight $\{\tau_1, \tau_2\}$ | $\{0.2, 0.3\}$ for MuJoCo and CityLearn<br>$\{0.9, 0.9\}$ for AntMaze |
| | Balance coefficient $\alpha$ | 0.4 |
| | Mild constraint factor $\lambda$ | 10 for MuJoCo and CityLearn<br>2.5 for AntMaze |

## Appendix H. Normalized Score Learning Curve

This section reports the normalized score learning curves of ISSP and baseline methods over one million training steps on D4RL and 0.2 million steps on NeoRL. The results illustrate the training dynamics and facilitate a comprehensive evaluation of convergence behavior, performance improvement, and learning stability.



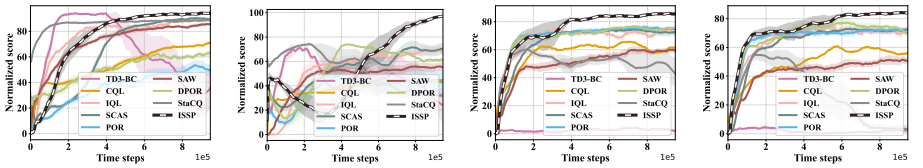

(a) halfcheetah-random (b) halfcheetah-medium (c) halfcheetah-med-rep (d) halfcheetah-med-exp

(e) halfcheetah-expert (f) hopper-random (g) hopper-medium (h) hopper-med-rep

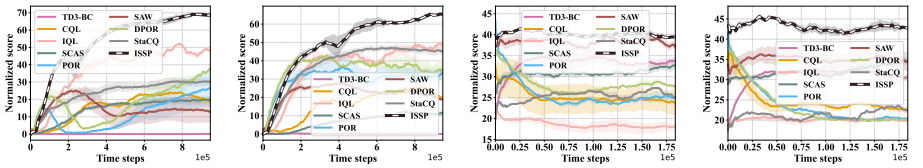
(i) hopper-med-exp (j) hopper-expert (k) walker2d-random (l) walker2d-medium

(m) walker2d-med-rep (n) walker2d-med-exp (o) walker2d-expert

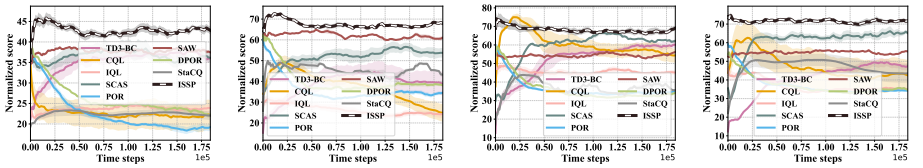
Fig. H3. Normalized score learning curves of ISSP against ORL baselines on the D4RL MuJoCo dataset tasks. The centerline and its shadow represent the mean and standard deviation across five seeds, respectively. Smoothing via a moving average window is applied to enhance visual clarity. medium-replay=med-rep, medium-expert=med-exp



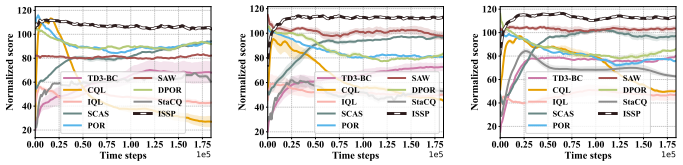

(a) antmaze-umaze (b) antmaze-umaze-diverse (c) antmaze-medium-play (d) antmaze-medium-diver




(e) antmaze-large-play (f) antmaze-large-diverse (g) citylearn-low-99 (h) citylearn-low-999




(i) citylearn-low-9999 (j) citylearn-medium-99 (k) citylearn-medium-999 (l) citylearn-medium-9999




(m) citylearn-high-99 (n) citylearn-high-999 (o) citylearn-high-999

Fig. H4. Normalized score learning curves of ISSP against ORL baselines on the D4RL AntMaze and NeoRL CityLearn dataset tasks.